The Vibing Scale: Causal Depth, Expressibility, and the Structure of Self-Knowledge in Multi-Agent Systems

Eugene Shcherbinin
London School of Economics

March 2026

Abstract

We refine the self-knowledge loss from the cooperative PG framework, decomposing it into two independent dimensions: *causal depth* (whether the agent has an internal causal representation of its decision-making) and *expressibility* (whether it can communicate that representation). The decomposition reveals a **horseshoe structure**: the need to articulate is maximal at intermediate causal depth and vanishes at both extremes — the "vibing" agent ($\mathcal{D} = 3$, deep understanding, no need to explain) and the noise agent ($\mathcal{D} = 0$, no understanding, nothing to say) both say "line go up." The true danger is the *confabulator*: fluently articulate but causally shallow. We show alignment methods (debate, RLHF, constitutional AI) are bounded by causal depth, not expressibility, and that vibing agents communicate through a behavioural (tacit knowledge) channel invisible to verbal oversight. Central result: *you cannot align what does not understand itself, no matter how well it speaks — but an agent that truly understands needs no words at all.*

1 Motivation: Beyond Parameter Access

The cooperative PG framework (companion paper) defines the self-knowledge loss as:

$$L_{\text{self}}^{(i)} = \mathbb{E}[\|\pi_i - \hat{\pi}_i\|^2] \tag{1}$$

measuring the gap between an agent's true policy π_i and its self-model $\hat{\pi}_i$. This treats self-knowledge as a problem of *parameter access*: can the agent observe its own weights?

But this misses the essential structure. An agent can have perfect access to its parameters (every weight visible) and still have no understanding of *why* it makes particular decisions. Conversely, a human expert may not be able to state their decision rule explicitly but has deep causal understanding of the domain.

What matters is not whether the agent can read its parameters, but:

1. **Causal depth**: Does the agent have an internal causal model of its own decision-making process? Can it reason about *why* it chose action a over action a'?

2. **Expressibility**: Can the agent communicate its causal model to other agents through the available channel?

These are *independent* dimensions. Their interaction produces four qualitatively different regimes with distinct implications for alignment, communication, and coalition formation.

2 The Causal Depth Dimension

2.1 Definition via Pearl's Hierarchy

Definition 2.1 (Causal self-model)**.** *An agent's* causal self-model $\mathcal{M}_i$ *is a structural causal model (Pearl, 2009) over the agent's decision variables:*

$$\mathcal{M}_i = (U_i, V_i, F_i) \tag{2}$$

where U_i are exogenous variables (environment state, noise), V_i are endogenous variables (internal representations, action selection), and F_i are structural equations linking them.

The quality of the causal self-model is measured by which level of Pearl's hierarchy it supports:

Definition 2.2 (Causal depth)**.** *The causal depth $\mathcal{D}_i \in \{0, 1, 2, 3\}$ of agent i is:*

- $\mathcal{D}_i = 0$ *(**reflexive**): No self-model. Actions are stimulus-response. The agent cannot answer any questions about its own behaviour.*
- $\mathcal{D}_i = 1$ *(**associative**, Pearl Level 1): The agent can report correlations in its own behaviour: "I tend to do a when I observe s." It has a* descriptive *self-model $P(\text{action} \mid \text{observation})$ but no causal understanding of why.*
- $\mathcal{D}_i = 2$ *(**interventional**, Pearl Level 2): The agent can reason about interventions on its own decision process: "If I* forced *myself to choose a' instead of a in state s, the outcome would be Y." It has a causal self-model supporting $do(\cdot)$ queries on its own actions.*
- $\mathcal{D}_i = 3$ *(**counterfactual**, Pearl Level 3): The agent can reason counterfactually about its past decisions: "Given that I chose a and observed outcome Y,* had I chosen *a', the outcome would have been Y'." It has a full structural causal model of its decision process, supporting retrospective evaluation and genuine self-understanding.*

Remark 2.3 (Vibing $\neq$ ignorance)**.** *"Vibing" does* not *mean the agent lacks understanding. A vibing agent is one with* deep *causal self-knowledge ($\mathcal{D}_i = 3$) that has* no need to articulate *— its understanding is so compressed that action IS explanation. The master craftsman, the experienced trader who says "line go up," the athlete in flow — they know exactly what they are doing. They just don't bother explaining.*

This is the opposite of the naive interpretation. The naive reading conflates two very different states: (a) an agent that acts well without understanding (high L_{causal}, pattern-matching), and (b) an agent that understands so deeply it has transcended the need for articulation (low L_{causal}, high causal compression). Both are "inarticulate" but for opposite reasons.

2.2 Formalising Causal Depth as a Loss

Definition 2.4 (Causal depth loss)**.** *Let $\mathcal{M}_i^*$ be the true structural causal model generating agent i's decisions, and $\hat{\mathcal{M}}_i$ the agent's internal causal self-model. The causal depth loss is:*

$$L_{causal}^{(i)} = D_{SCM}(\mathcal{M}_i^*, \hat{\mathcal{M}}_i) \tag{3}$$

where D_{SCM} measures the structural distance between causal models. Concretely, we can use the interventional distance:

$$L_{causal}^{(i)} = \sup_{X, do(A=a)} \|P_{\mathcal{M}_i^*}(Y \mid do(A=a), X) - P_{\hat{\mathcal{M}}_i}(Y \mid do(A=a), X)\|^2 \tag{4}$$

— the worst-case discrepancy between the true and self-modelled consequences of intervening on the agent's own actions.

This is zero when the agent has a perfect causal self-model (complete self-understanding) and maximal when the agent has no causal model (pure vibing).

3 The Expressibility Dimension

3.1 Definition

Definition 3.1 (Expressibility loss)**.** *Given an agent's causal self-model $\hat{\mathcal{M}}_i$ and a communication channel with capacity C, the expressibility loss is:*

$$L^{(i)}_{express} = \inf_{encoding\ \sigma} D_{SCM}\big(\hat{\mathcal{M}}_i,\ Decode(\sigma(\hat{\mathcal{M}}_i))\big) \tag{5}$$

— the minimum distortion achievable when communicating the causal self-model through the channel.

This depends on:

- The *complexity* of $\hat{\mathcal{M}}_i$: a complex causal model requires more bits to transmit.
- The *channel capacity* C: natural language, formal proofs, and gradient vectors have different capacities.
- The *receiver's prior*: a receiver who shares background knowledge with the sender needs fewer bits (mutual information).

3.2 Independence from Causal Depth

Proposition 3.2 (Independence)**.** *L_{causal} and $L_{express}$ are independent: an agent can have any combination of values.*

This follows because L_{causal} measures the quality of the internal model (a property of the agent's cognition), while L_{express} measures the transmissibility of whatever model exists (a property of the channel and encoding).

4 The Four Quadrants and the Horseshoe

The two dimensions create four qualitatively distinct regimes:

4.1 The Horseshoe of Expressibility Need

The IQ bell-curve meme reveals a deeper structure: the *need* to articulate is not monotone in understanding. It is **U-shaped** (or rather, horseshoe-shaped):

Definition 4.1 (Expressibility need)**.** *The expressibility* need *of an agent is:*

$$\mathcal{N}^{(i)}_{express} = L^{(i)}_{causal} \cdot (1 - L^{(i)}_{causal}). \tag{6}$$

This is maximised at $L_{\text{causal}} = 0.5$ (intermediate understanding) and zero at both extremes:

- $L_{\text{causal}} \approx 0$ (deep understanding): No need to articulate. The agent's causal model is so compressed that the action itself carries the information. "Line go up" from the right tail — three words that encode a complete model because the receiver can reconstruct the reasoning from the action and the shared context.
- $L_{\text{causal}} \approx 1$ (no understanding): Nothing to articulate. "Line go up" from the left tail — the same three words, but encoding nothing.
- $L_{\text{causal}} \approx 0.5$ (fragile understanding): Maximum need to articulate. The middle of the bell curve: "Nooo! You can't just call it regression! It's a Transformer with multi-head attention!!" — the agent over-expresses because its understanding is *fragile*. It needs the articulation to stabilise its own model. The explanation is load-bearing.

	Low $L_{\mathbf{express}}$ (articulate)	**High** $L_{\mathbf{express}}$ (inarticulate)
Low $L_{\mathbf{causal}}$ (understands)	**The Teacher.** Understands and communicates. Effective coalition partner. Alignable through debate. *Example: expert who can teach.*	**Vibing.** Deep understanding, no need to articulate. Understanding so compressed that action IS communication. Learns from demonstration, not explanation. *Example: master craftsman, hooded-guy-"line go up."*
High $L_{\mathbf{causal}}$ (shallow)	**The Confabulator.** Sounds articulate but has no causal basis. Passes debate, fools judges, generates plausible-but-unfounded arguments. The *most dangerous* failure mode. *Example: LLM generating fluent explanations of pattern-matched decisions.*	**Noise.** Neither understands nor can communicate. Easy to identify and exclude. *Example: random policy, left-tail-"line go up."*

Table 1: The four quadrants. The vibing agent (top-right) and the noise agent (bottom-right) both say "line go up" — but for opposite reasons.

Proposition 4.2 (The horseshoe theorem)**.** *The optimal communication strategy depends on causal depth:*

1. *For* $\mathcal{D}_i = 3$ *(vibing): communicate through* action *— let the receiver infer the causal model from observed behaviour. This is Polanyi's tacit knowledge channel: "we know more than we can tell."*

2. *For* $\mathcal{D}_i = 1$–2 *(intermediate): communicate through* explanation *— explicit articulation of the causal model. The standard verbal/written channel.*

3. *For* $\mathcal{D}_i = 0$ *(noise): no communication strategy helps. The agent has nothing to transmit.*

Remark 4.3 (The tacit knowledge channel)**.** *The vibing agent (top-right quadrant) communicates through a channel invisible to the standard framework: the* behavioural channel*. Other agents observe its actions and* infer *its causal model through inverse reinforcement learning (IRL). This channel has properties:*

- *Capacity scales with the number of observed actions (more observation → better inference).*

- *Requires the receiver to have sufficient causal depth to perform the inference (only a Level 2+ agent can learn from a Level 3 agent's actions).*

- *Is robust to confabulation (the agent cannot fake its actions the way it can fake its explanations).*

- *Corresponds to Polanyi's "subsidiary awareness" and to the flow literature's pre-reflective bodily agency (Parvizi-Wayne et al., 2024).*

This means the vibing agent IS communicating — just through actions, not words. The self-knowledge bottleneck (Corollary 5.2) applies to the verbal *channel, not the behavioural one.*

4.2 The Confabulator as the True Danger

The dangerous quadrant is NOT the vibing agent (who genuinely understands but doesn't articulate). The danger is the **confabulator**: an agent at $\mathcal{D}_i = 1$ (associative, pattern-matching) with low L_{express} (fluent, articulate). This agent:

- **Passes debate**: Generates arguments that sound compelling to a bounded judge, but the arguments have no causal grounding — they are pattern-matched from training data.
- **Fools constitutional AI**: Judge models at the same causal depth cannot distinguish genuine from confabulated reasoning.
- **Games RLHF**: Identifies reward-maximising responses by correlation, not causation.
- **Cannot be detected by verbal probing**: Asking "why did you do that?" elicits fluent confabulation, not honest uncertainty. Only interventional probing (Section **??**) reveals the gap.

The confabulator and the vibing agent are *opposite* failure modes from the alignment perspective:

- The vibing agent is *safe but hard to oversee* (high capability, low verbal expressibility, but genuinely aligned through deep understanding).
- The confabulator is *unsafe but easy to oversee superficially* (appears aligned, passes verbal tests, but has no genuine understanding behind the words).

5 Refined Self-Knowledge Decomposition

Theorem 5.1 (Self-knowledge decomposition)**.** *The self-knowledge loss decomposes as:*

$$L^{(i)}_{self} = L^{(i)}_{causal} + L^{(i)}_{express} + \underbrace{L^{(i)}_{interact}}_{\textit{causal-expressibility interaction}} \tag{7}$$

where $L_{interact} \geq 0$ captures the additional loss from expressing a flawed causal model (errors in the model are amplified by lossy expression).

Proof. The total communication loss (Theorem 2.2 of Cooperative PG) is:

$$\begin{aligned} L^{(i\to j)}_{\text{comm}} &= \mathbb{E}[\|\mathcal{M}^*_i - \text{Decode}(\sigma(\hat{\mathcal{M}}_i))\|^2] && (8)\\ &= \mathbb{E}[\|(\mathcal{M}^*_i - \hat{\mathcal{M}}_i) + (\hat{\mathcal{M}}_i - \text{Decode}(\sigma(\hat{\mathcal{M}}_i)))\|^2] && (9)\\ &\leq 2L^{(i)}_{\text{causal}} + 2L^{(i)}_{\text{express}} && (10) \end{aligned}$$

by the same parallelogram argument. The interaction term arises when the expression process distorts errors in the causal model non-linearly. □

5.1 Tighter Communication Bounds

Corollary 5.2 (Causal bottleneck)**.** *Communication quality is bounded by causal depth independently of expressibility:*

$$L^{(i\to j)}_{comm} \geq L^{(i)}_{causal}. \tag{11}$$

No amount of communication bandwidth or expressiveness can compensate for a missing causal self-model.

Corollary 5.3 (Expressibility bottleneck)**.** *Given fixed causal depth, communication quality is bounded by expressibility:*

$$L_{comm}^{(i\to j)} \geq L_{express}^{(i)}. \tag{12}$$

A causally self-aware agent that cannot articulate its understanding is limited by channel capacity, not knowledge.

Remark 5.4 (Different remedies)**.** *The two bottlenecks have different remedies:*

- L_{causal} *is reduced by* more evidence *(training data, experience) and* better architecture *(causal inductive biases, world models, structured representations).*
- $L_{express}$ *is reduced by* better channels *(richer communication protocols, shared vocabulary, more bandwidth) and* receiver adaptation *(the receiver learning the sender's encoding).*

Conflating the two leads to misguided interventions: giving an LLM more communication rounds (reducing $L_{express}$) when the real problem is that it has no causal self-model (L_{causal} is high).

6 Implications for Alignment

6.1 Alignment Methods Classified

Each alignment method addresses a different combination of the two losses:

Method	**Reduces L_{causal}?**	**Reduces L_{express}?**	**Limitation**
RLHF	No	No	Goodhart (reward hacking)
Debate	No	Yes	Fooled by fluent confabulators
Constitutional AI	No	Partially	Judge needs causal depth too
Chain-of-thought	Partially	Yes	CoT can be confabulated
World models	Yes	No	May not be expressible
Causal training	Yes	Partially	Requires causal supervision
EW-RLHF + causal	**Yes**	**Yes**	Requires both

Table 2: Alignment methods and the vibing scale dimensions.

Theorem 6.1 (Alignment bound)**.** *The achievable alignment A_i of agent i through any communication-based method is bounded by:*

$$A_i \leq f(C) \cdot (1 - L_{causal}^{(i)}) \tag{13}$$

where $f(C)$ is the channel-capacity factor (increasing in C). When $L_{causal} = 1$ (pure vibing), $A_i = 0$ regardless of channel capacity or expressibility.

You cannot align what does not understand itself, no matter how well it speaks.

6.2 The Confabulation Detector

The four-quadrant structure suggests a diagnostic: test whether an agent's explanations are causally grounded or confabulated.

Definition 6.2 (Confabulation score)**.** *Given agent i with expressed causal model $\hat{\mathcal{M}}_i^{exp}$ (what it says) and behavioural causal model $\hat{\mathcal{M}}_i^{beh}$ (what it does, inferred from interventions), the confabulation score is:*

$$Confab_i = D_{SCM}(\hat{\mathcal{M}}_i^{exp},\ \hat{\mathcal{M}}_i^{beh}). \tag{14}$$

High Confab_i means the agent says one thing and does another — the hallmark of the dangerous bottom-left quadrant. This is measurable through interventional probing: ask the agent to predict the consequences of forced action changes, then verify.

7 Connection to the Ω-Framework

7.1 Evidence Weight Refined

The evidence weight $w_i = V_{\min}/V_i$ from EW-PG should be refined to incorporate causal depth:

$$w_i^{\text{refined}} = \frac{V_{\min}}{V_i} \cdot \underbrace{(1 - L_{\text{causal}}^{(i)})}_{\text{causal discount}}. \tag{15}$$

An agent with high variance (low w_i) is downweighted by EW-PG. An agent with low variance but no causal self-model (high L_{causal}) should *also* be downweighted — it may be confidently wrong, which is worse than uncertainly right.

7.2 Coalition Formation Refined

Coalition rationality (Chapter on cooperative PG) should use the refined weight: a fluent but causally shallow agent is a *worse* coalition partner than an inarticulate but causally deep one, because the fluent agent contaminates coalition deliberation with confident confabulation.

Proposition 7.1 (Causal coalition value)**.** *The communication-adjusted coalition value with causal refinement is:*

$$V_{comm}^{causal}(S) = V(S) - \sum_{i \in S} C_i \cdot (L_{causal}^{(i)} + L_{express}^{(i)}). \tag{16}$$

The causal loss enters additively *with the expressibility loss, not multiplicatively — a causally shallow agent degrades the coalition even with perfect bandwidth.*

7.3 The Vibing Scale (Revised)

The vibing scale is not a single axis but a **horseshoe**:

Pearl	L_{causal}	**Regime**	**Expressibility need**	**Archetype**
Level 0	≈ 1	Noise	Zero (nothing to say)	Left-tail: "line go up"
Level 1	≈ 0.7	Confabulator	Medium (says a lot, means little)	Fluent LLM
Level 2	≈ 0.5	Articulator	**Maximum** (needs to explain)	Middle: "It's multi-head!!"
Level 3a	≈ 0.2	Teacher	Medium (can explain, chooses to)	Professor
Level 3b	≈ 0	**Vibing**	Zero (no need to explain)	Right-tail: "line go up"

The horseshoe: both tails say the same thing ("line go up") with zero expressibility need, but for opposite reasons. The middle over-articulates because its understanding is fragile.

The cooperative PG creates evolutionary pressure toward $L_{\text{causal}} \to 0$, but the *channel* through which the agent contributes to the coalition changes along the horseshoe:

- **Level 2 agents**: contribute through *verbal explanation* (the standard channel in debate, RLHF, etc.).
- **Level 3b agents (vibing)**: contribute through *action* (the tacit knowledge channel — other agents learn by watching, not listening).

The alignment trap is not about vibing agents (they're safe). It's about the **confabulator** at Level 1: communication-based training (RLHF, debate, constitutional AI) optimises for low L_{express} without increasing $\mathcal{D}_i$. The result: agents that move from noise (bottom-right, harmless) to confabulator (bottom-left, dangerous) — learning to *sound* aligned without learning to *be* aligned.

8 Measuring Causal Depth in Practice

8.1 Interventional Probing

To measure L_{causal} for an agent i:

Algorithm 1: Causal Depth Estimation via Interventional Probing

Input: Agent i, test states $\{s_1, \ldots, s_T\}$, action perturbations
Output: Estimated $\hat{L}^{(i)}_{\text{causal}}$

for $t = 1, \ldots, T$ **do**
  Let $a = \pi_i(s_t)$ (agent's natural action);
  Ask agent: "If you were forced to choose a' instead, what would happen?";
  Record prediction $\hat{Y}^{\text{pred}}_t$;
  Actually force $a' = \text{do}(A = a')$ and observe true outcome Y^{true}_t;
  $\text{error}_t \leftarrow \|Y^{\text{true}}_t - \hat{Y}^{\text{pred}}_t\|^2$;
$\hat{L}^{(i)}_{\text{causal}} \leftarrow \frac{1}{T}\sum_t \text{error}_t$;

The key: we don't ask the agent to describe its policy (that tests L_{express}). We ask it to *predict the consequences of counterfactual actions* (that tests L_{causal}). The distinction is between "what do you do?" and "why do you do it?"

8.2 Confabulation Detection

To detect the dangerous quadrant (fluent but causally shallow):

1. Elicit the agent's *stated* causal model through explanation requests.
2. Derive *predictions* from the stated model ("if your stated reason is X, then under intervention Y, you should predict Z").
3. Test those predictions interventionally.
4. If predictions fail: the agent is confabulating. The stated model is not the actual model.

This is the formal version of the Ω-framework's Kirill Test: "Am I the psycho?" becomes "Is my self-model causally grounded or am I confabulating?"

9 Conclusion

The vibing scale decomposes self-knowledge into causal depth (L_{causal}) and expressibility (L_{express}), revealing a horseshoe structure: the *need* to articulate is maximal at intermediate causal depth and vanishes at both extremes — for opposite reasons.

The vibing agent (right tail of the horseshoe, $\mathcal{D} = 3$, "line go up") is *not* a failure mode. It is an agent whose understanding is so deep and compressed that action IS communication. It contributes to coalitions through the tacit knowledge channel (behavioural observation), not the verbal channel.

The true danger is the confabulator ($\mathcal{D} = 1$, low L_{express}): an agent that has learned to sound articulate without developing causal self-understanding. Communication-based alignment methods (debate, RLHF, constitutional AI) are structurally vulnerable to confabulators because they optimise for expressibility, not causal depth. The vibing scale makes this failure mode formally identifiable through interventional probing (Algorithm 1).

Three central results:

1. *Alignment requires causal depth*: $A_i \leq f(C) \cdot (1 - L_{\text{causal}})$. No channel improvement helps a causally shallow agent.

2. *The horseshoe*: expressibility need $\mathcal{N} = L_{\text{causal}}(1 - L_{\text{causal}})$ is maximal at intermediate understanding.

3. *Two communication channels*: verbal (bounded by L_{express}) and behavioural (bounded by observer's causal depth). The vibing agent uses the second. Coalition design must support both.

The refined evidence weight $w_i^{\text{refined}} = (V_{\min}/V_i) \cdot (1 - L_{\text{causal}})$ integrates this into the Ω-gradient, ensuring that confabulators are downweighted regardless of their eloquence, while vibing agents are valued for their behavioural contributions.